DTU Health Tech
Department of Health Technology

Deep Learning Approaches for Cancer of Unknown Primary (CUP) Classification

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M.Sc. Thesis
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Thesis in collaboration with Rigshospitalet, MDxCore Department

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Abstract

Cancer of unknown primary (CUP) represents an aggressive and heterogeneous group of metastatic cancers in which the tissue of origin cannot be identified through standard diagnostic work-up, limiting the use of site-specific treatment strategies. This thesis investigates whether RNA-seq-based representation learning can support tissue-of-origin prediction in CUP-like samples.

A variational autoencoder (VAE), a generative model that learns a probabilistic low-dimensional latent space, is used to obtain compressed representations of RNA-seq profiles. Using a reference transcriptomic dataset containing both tumor and normal samples, three classification strategies are compared: multinomial logistic regression trained on raw gene expression data, VAE-derived latent embeddings, and embeddings from a VAE regularized with supervised contrastive loss. Model performance was evaluated on a held-out internal test set and an independent cohort of metastatic biopsies from Rigshospitalet.

In the internal evaluation, all models achieved accuracy above 97%, indicating that RNA-seq profiles contain a strong tissue-of-origin signal. However, performance decreased substantially in the external metastatic cohort, with misclassifications mainly occurring among transcriptionally related tumor types. These findings suggest that contrastive representation learning can improve cross-cohort generalization, while also highlighting the gap between internal benchmark performance and clinical validation. Overall, this work supports the potential of RNA-seq-based machine learning approaches for CUP classification, but emphasizes the need for broader reference cohorts, multimodal integration, and careful clinical interpretation before clinical implementation.